

11 Particle physics

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Hadrons are made of quarks: baryons contain three quarks and mesons a quark–antiquark pair. Leptons, including electrons and neutrinos, are fundamental. Particle reactions must conserve charge, baryon number and lepton number; mass–energy may be converted according to $E = mc^2$.

Worked example: follow the reasoning

A proton has quarks uud: charges $+2/3 + 2/3 - 1/3 = +1$ in elementary charge units. A neutron udd has total charge zero.

Quick check · multiple choice

A What is the quark content of a proton?

[1]

- A udd
- B uuu
- C uud
- D ddd

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 In β^- decay, a neutron becomes a proton, electron and electron antineutrino. Verify charge and lepton-number conservation.

[3]

Answers and explanation

Quick check: C. uud

Question 1 [3 marks]

Charge: $0 \rightarrow +1 -1 +0 = 0$ [1]. Electron lepton number: $0 \rightarrow +1 -1 = 0$ [1]. Baryon number remains +1 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.